

OpenWorldLib: One Framework to Define and Run Every World Model

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The banner features a dark blue background with a subtle grid pattern. At the top, the title 'OpenWorldLib: One Framework For Every World Model' is displayed in large, bold, white and teal text. Below the title, a subtitle reads 'A unified definition and codebase from Peking University'. The main content is organized into three vertical columns, each with a colored header and descriptive text. The first column is titled '5 Modules' in orange and lists 'Operator, synthesis, reasoning, representation, memory'. The second column is titled '13+ Models' in yellow and lists 'Hunyuan, Cosmos, VGGT, πo, Matrix-Game and more'. The third column is titled '680 Stars' in blue and lists 'Apache 2.0 open source, 19 contributors'. At the bottom left, the date 'April 19, 2026' is shown, and at the bottom right, the 'ToKnow.ai' logo is present.

OpenWorldLib: One Framework For Every World Model

A unified definition and codebase from Peking University

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The Peking University DataFlow team released [OpenWorldLib](#), a unified codebase and a sharper definition for what a “world model” actually is. The term has been used for everything from next-frame video predictors to robot policies, which makes fair comparison nearly impossible. The paper proposes a single working definition: a world model is a system centered on perception, equipped with interaction and long-term memory, that understands

and predicts a complex world. The framework breaks that into five modules: an operator that handles input and interaction signals, a synthesis module for video, 3D, and robot actions, a reasoning module for spatial and multimodal reasoning, a representation module for 3D structure, and a memory module for long-horizon context. The [GitHub repository](#) ships under Apache 2.0 and already integrates 13 systems including Matrix-Game-2, Hunyuan-WorldPlay, Cosmos-Predict-2.5, WoW, VGGT, and the `llm-vlm` and `llm-vlm-act` vision-language-action models, with 680 stars at the time of writing.

For anyone building or evaluating world models, this removes a real source of friction. Instead of cloning a different repo for each method and writing custom glue code to compare interactive video generation against 3D scene reconstruction or robot action prediction, you load a shared pipeline and call standardized methods. The paper’s evaluation across these systems also surfaces a useful state-of-the-art map: Hunyuan-WorldPlay leads on visual quality for interactive video, VGGT can build 3D scenes but drifts geometrically under large camera motion, and `llm-vlm` generalizes well across robot tasks but long-horizon physical consistency remains an open problem across the board.

The bigger shift is that world model research is starting to look less like one-off papers and more like a stack with shared interfaces. Once memory, perception, and action generation can be swapped independently, progress can compound across labs instead of restarting at each release.

Read More: [Trinity of Consistency: a framework for measuring general world models](#)

Sources:

- [OpenWorldLib: A Unified Codebase and Definition of Advanced World Models \(arXiv\)](#)
- [OpenWorldLib GitHub repository](#)
- [OpenWorldLib HTML paper](#)
- [DataFlow project \(parent codebase\)](#)

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